Software Requirements Specification

Newspaper Agency Automation Software

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Revision History

Version	Date	Description	Author
1.0	March 2026	Initial release of SRS document	SE Team

Contents

Chapter 1

Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the **Newspaper Agency Automation Software (NAAS)**. The system is being developed at the request of a local newspaper and magazine delivery agency to automate various clerical and operational activities associated with its business.

The intended audience for this document includes:

- Project managers and developers responsible for system design and implementation.
- Quality assurance engineers responsible for testing and validation.
- The client (agency manager) for review and approval.
- Maintenance and support personnel.

1.2 Scope

The **Newspaper Agency Automation Software** is a desktop/web-based information system designed to manage the end-to-end operations of a newspaper and magazine delivery agency. The system will:

- Automate generation of daily delivery lists for each delivery person.
- Manage customer subscriptions (new subscriptions, modifications, and cancellations).
- Generate monthly bills for each customer automatically.
- Handle payment recording (cash and cheque) and receipt generation.
- Track outstanding dues and issue reminder notices.
- Calculate and print delivery person commission reports.
- Provide the manager with summary reports on publications delivered.

The system does **not** include:

- Online/e-commerce customer portal.

- Integration with external banking or payment gateways.
- Inventory management for publication stocks.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
NAAS	Newspaper Agency Automation Software
Manager	The owner or administrator of the newspaper delivery agency
Delivery Person	Field employee responsible for delivering publications
Customer	A subscriber who receives one or more publications
Subscription	A formal agreement by a customer to receive publications
Publication	Any newspaper or magazine delivered by the agency
Billing Period	A calendar month for which bills are generated
Outstanding Due	Unpaid amount from a previous billing period
Commission	2.5% of the total value of publications delivered by a delivery person
IEEE	Institute of Electrical and Electronics Engineers
UI	User Interface
DB	Database
DBMS	Database Management System

1.4 References

1. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Computer Society, 1998.
2. Client Requirements Document (informal brief), Newspaper Agency, February 2026.
3. Project Charter – NAAS Project, February 2026.

1.5 Overview

The remainder of this SRS is structured as follows:

- **Chapter 2** provides an overall description of the product including its context, major functions, user classes, constraints, and assumptions.

- **Chapter 3** presents the specific requirements covering external interfaces, functional requirements, performance requirements, database design, design constraints, quality attributes, and object-oriented models.
- **Chapter 4** contains appendices with supporting material.
- **Chapter 5** provides an index for quick reference.

Chapter 2

Overall Description

2.1 Product Perspective

The Newspaper Agency Automation Software is a new, standalone system. It replaces the manual, paper-based record-keeping currently used by the agency. The system interacts with:

- **Users:** The agency manager and delivery persons.
- **Printer:** A locally connected printer for generating delivery lists, bills, receipts, reminder notices, and commission reports.
- **Database:** A local relational database storing customer, subscription, publication, payment, and delivery data.

The system is self-contained and does not require integration with external systems at this stage. A high-level context diagram is shown conceptually below:

[Manager / Delivery Person] $\longleftrightarrow$ *[NAAS System]* $\longleftrightarrow$ *[Database & Printer]*

2.2 Product Functions

The major functions of the NAAS are:

1. **Subscription Management** – Add, modify, and cancel customer subscriptions with mandatory one-week advance notice.
2. **Delivery List Generation** – Print daily delivery lists per delivery person, ordered by address to minimise travel distance.
3. **Delivery Hold Management** – Record and process temporary delivery stops for customers who are out of station.
4. **Monthly Billing** – Automatically compute and print itemised bills at the beginning of each month.
5. **Payment Processing** – Record cash or cheque payments and print receipts.
6. **Outstanding Due Tracking** – Send polite reminders for dues outstanding more

than one month; suspend subscriptions after two months of non-payment.

7. **Manager Reporting** – Print summary reports of publications delivered and customer-wise delivery details for the current month.

8. **Commission Calculation** – Calculate and print the commission payable to each delivery person (2.5% of delivered publication value).

2.3 User Characteristics

User Class	Characteristics
Agency Manager	Primary administrator of the system. Has full access to all features. Expected to have basic computer literacy. Responsible for subscription management, billing, payment entry, and report generation.
Delivery Person	Field operator. Has limited system access restricted to viewing and printing their own daily delivery lists. May have limited computer skills; the interface must be simple and intuitive.

2.4 Constraints

1. **Subscription Change Notice:** The system must enforce a minimum one-week advance notice period for all subscription changes.

2. **Payment Methods:** Only cash and cheque payment modes are supported; online payments are out of scope.

3. **Bill Generation Timing:** Bills must be generated automatically at the beginning (first day) of each calendar month.

4. **Commission Rate:** Delivery person commission is fixed at 2.5% of the total publication value delivered and cannot be configured by end users.

5. **Suspension Policy:** Subscriptions *must* be suspended automatically after two consecutive months of outstanding dues with no exceptions.

6. **Hardware:** The system must run on standard desktop hardware with a locally attached printer.

7. **Regulatory:** The system must maintain records for a minimum of three years for audit purposes.

2.5 Assumptions and Dependencies

1. Customers are assigned unique IDs at the time of registration and are assumed to have a fixed primary delivery address.
2. Publication prices are assumed to be set by the manager and stored in the system; price changes are applied from the next billing cycle.
3. The system assumes reliable access to a local printer during report and bill generation.
4. Delivery route optimisation is based on consecutive address ordering rather than a full geo-spatial algorithm (i.e., addresses are sorted lexicographically or by a pre-assigned route sequence).
5. The agency operates on calendar months; partial-month deliveries are prorated.
6. A stable relational DBMS (e.g., MySQL, PostgreSQL, or SQLite) is available on the deployment machine.

Chapter 3

Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

- The system shall provide a **graphical user interface (GUI)** accessible via a desktop application or web browser.
- The manager's interface shall include menus for: Customer Management, Subscription Management, Billing, Payments, Reports, and Delivery Person Management.
- The delivery person's interface shall be restricted to viewing and printing their assigned daily delivery list.
- All data entry forms shall include inline validation with descriptive error messages.
- The system shall support both keyboard navigation and mouse interaction.

3.1.2 Hardware Interfaces

- The system shall interface with a standard locally connected printer (USB or network) for printing delivery lists, bills, receipts, and reports.
- The system shall run on a standard PC with minimum 4 GB RAM, 2 GHz processor, and 20 GB free disk space.

3.1.3 Software Interfaces

- The system shall use a relational DBMS (MySQL 8.x / PostgreSQL 14+ / SQLite 3.x) via a standard database driver/ORM.
- Report and bill documents shall be generated as PDF files for printing.
- The system shall be deployable on Windows 10+ or Ubuntu 20.04+.

3.1.4 Communication Interfaces

- No external network communication is required in the initial version.
- All data communication is local (application $\leftrightarrow$ local database).

3.2 Functions

The following table enumerates all functional requirements. Each requirement is identified by a unique ID.

Req. ID	Requirement Description	Priority
FR-01 – Customer Management		
FR-01.1	The system shall allow the manager to register a new customer with name, address, contact details, and a unique customer ID.	High
FR-01.2	The system shall allow the manager to update customer details.	High
FR-01.3	The system shall allow the manager to deactivate a customer account upon request for permanent cancellation.	Medium
FR-02 – Subscription Management		
FR-02.1	The system shall allow a customer to subscribe to one or more publications.	High
FR-02.2	The system shall allow customers to modify their subscription list with a minimum of one week's advance notice.	High
FR-02.3	The system shall allow customers to discontinue their entire subscription.	High
FR-02.4	The system shall enforce the one-week advance notice rule and reject changes that do not comply, displaying an appropriate message.	High
FR-02.5	The system shall maintain a history of all subscription changes with timestamps.	Medium
FR-03 – Delivery Hold Management		
FR-03.1	The system shall allow customers to request a temporary stop on deliveries for a specified date range.	High
FR-03.2	The system shall automatically resume deliveries when the hold period expires.	High
FR-03.3	Deliveries suspended during a hold period shall not be billed for that period.	High

Req. ID	Requirement Description	Priority
FR-04 – Daily Delivery List Generation		
FR-04.1	The system shall generate a daily delivery list for each delivery person every day.	High
FR-04.2	The delivery list shall show, for each address: customer name, address, and the list of publications to be delivered (name and number of copies).	High
FR-04.3	Addresses on the delivery list shall be ordered consecutively to minimise the delivery person's travel.	High
FR-04.4	The system shall exclude customers on a delivery hold from the list.	High
FR-04.5	The delivery list shall be printable as a formatted document.	High
FR-05 – Monthly Billing		
FR-05.1	The system shall automatically generate bills for all active customers at the beginning of each month.	High
FR-05.2	Each bill shall include: publication name, number of copies delivered during the month, unit price, and total cost per publication.	High
FR-05.3	Each bill shall include any outstanding dues from previous months.	High
FR-05.4	Bills shall be printable and ready for delivery to customers.	High
FR-06 – Payment Processing		
FR-06.1	The system shall allow the manager to record a payment as either cash or cheque, along with amount and date.	High
FR-06.2	For cheque payments, the system shall record the cheque number.	High
FR-06.3	Upon recording a payment, the system shall automatically print a receipt for the customer.	High

Req. ID	Requirement Description	Priority
FR-06.4	The system shall update the customer's outstanding balance upon payment entry.	High
FR-07 – Outstanding Dues and Suspension		
FR-07.1	The system shall track outstanding dues for each customer on a monthly basis.	High
FR-07.2	If a customer's dues are outstanding for more than one month, the system shall print a polite reminder notice.	High
FR-07.3	If a customer's dues remain outstanding for more than two months, the system shall automatically suspend their subscription.	High
FR-07.4	The system shall notify the manager of all suspended subscriptions.	Medium
FR-07.5	Suspended subscriptions shall be reinstated automatically once full payment is received.	Medium
FR-08 – Manager Reports		
FR-08.1	The system shall generate a monthly summary report showing total publications delivered, grouped by publication type.	High
FR-08.2	The system shall generate a detailed report showing which customer received which publications during the current month.	High
FR-08.3	All reports shall be printable as formatted documents.	High
FR-09 – Delivery Person Commission		
FR-09.1	The system shall calculate the commission payable to each delivery person as 2.5% of the total value of publications delivered by that person.	High
FR-09.2	The system shall print a commission statement for each delivery person at the end of each month.	High
FR-09.3	The commission statement shall detail publications delivered, total value, and computed commission amount.	Medium

3.3 Performance Requirements

1. **Response Time:** All interactive operations (form submissions, queries) shall complete within 3 seconds under normal load (1–5 concurrent users).
2. **Batch Processing:** Monthly bill generation for up to 1,000 customers shall complete within 5 minutes.
3. **Delivery List Generation:** Daily delivery lists for all delivery persons shall be generated within 2 minutes.
4. **Print Output:** A bill or receipt document shall be spooled to the printer within 10 seconds of the manager's request.
5. **Data Volume:** The system shall support up to 1,000 active customers, 50 publications, and 20 delivery persons without performance degradation.
6. **Availability:** The system shall be available during business hours (6 AM – 10 PM) with a target uptime of 99.5%.

3.4 Logical Database Requirements

The following entities and relationships form the logical data model:

3.4.1 Key Entities

Entity	Key Attributes
Customer	CustomerID (PK), Name, Address, Phone, Email, Status (Active/Suspended/Inactive)
Publication	PublicationID (PK), Name, Type (Newspaper/Magazine), UnitPrice, Frequency
Subscription	SubscriptionID (PK), CustomerID (FK), PublicationID (FK), StartDate, EndDate, NoOfCopies, Status
DeliveryPerson	DeliveryPersonID (PK), Name, AssignedZone, ContactNo
DeliveryRecord	RecordID (PK), CustomerID (FK), PublicationID (FK), DeliveryPersonID (FK), Date, CopiesDelivered
DeliveryHold	HoldID (PK), CustomerID (FK), StartDate, EndDate
Bill	BillID (PK), CustomerID (FK), BillingMonth, TotalAmount, OutstandingDue, GeneratedDate

Entity	Key Attributes
BillDetail	DetailID (PK), BillID (FK), PublicationID (FK), CopiesDelivered, CostAmount
Payment	PaymentID (PK), CustomerID (FK), Amount, Date, Mode (Cash/Cheque), ChequeNo, ReceiptNo
Commission	CommissionID (PK), DeliveryPersonID (FK), Month, TotalValue, CommissionAmount

3.4.2 Key Relationships

- A **Customer** may have **many Subscriptions** (one per publication).
- A **Subscription** links a Customer to a Publication.
- A **DeliveryPerson** is associated with many **DeliveryRecords**.
- A **Bill** has one or more **BillDetail** rows (one per publication).
- A **Customer** may have many **Payments** and many **Bills**.
- A **Customer** may have many **DeliveryHolds** over time.

3.5 Design Constraints

1. **Platform:** The system must run on Windows 10+ or Ubuntu 20.04+ without requiring cloud infrastructure.
2. **Database:** A standard relational DBMS must be used. NoSQL databases are not permitted.
3. **Print Format:** All printable outputs (bills, receipts, delivery lists) must be generated as PDF documents compatible with standard office printers.
4. **Data Retention:** The system must retain all financial and delivery records for a minimum of three years.
5. **Backup:** The system must support daily automated database backups to a local directory.
6. **Language:** The system UI shall be in English. Date formats shall follow DD/MM/YYYY.

3.6 Software System Quality Attributes

Attribute	Requirement
Reliability	The system shall not lose data in the event of an unexpected shutdown; all committed transactions must be durable (ACID compliance).
Usability	A new manager with basic computer skills shall be able to complete key tasks (entering a payment, generating a delivery list) within 30 minutes of initial training.
Security	The system shall require username/password login. The manager and delivery person roles shall have separate access privileges. Passwords shall be stored as salted hashes.
Maintainability	The system source code shall be modular and documented with inline comments. New publications shall be addable without code changes.
Portability	The application shall be deployable on Windows and Linux with no source code changes, only configuration differences.
Scalability	The database design shall support growth to 5,000+ customers without architectural changes.
Accuracy	All financial calculations (billing amounts, commission) shall be accurate to two decimal places with no rounding errors across the monthly cycle.

3.7 Object-Oriented Models

3.7.1 Organization of Section 3

Section 3 is organised around the major subsystems of NAAS: *Customer & Subscription Management*, *Delivery Management*, *Billing & Payment*, and *Reporting*. Object-oriented models describe the key classes and their interactions within these subsystems.

3.7.2 Key Classes (Conceptual)

- **Customer:** Attributes: customerID, name, address, phone, status. Methods: *subscribe()*, *cancelSubscription()*, *requestHold()*, *makePayment()*.
- **Subscription:** Attributes: subscriptionID, customer, publication, copies, startDate, endDate, status. Methods: *activate()*, *suspend()*, *modify()*.
- **Publication:** Attributes: publicationID, name, type, price. Methods: *getPrice()*,

updatePrice().

- **DeliveryPerson:** Attributes: deliveryPersonID, name, zone. Methods: *generateDailyList()*, *getCommission()*.
- **DeliveryList:** Attributes: date, deliveryPerson, deliveryItems[]. Methods: *generate()*, *print()*, *sortByAddress()*.
- **Bill:** Attributes: billID, customer, billingMonth, details[], totalAmount, outstandingDue. Methods: *compute()*, *print()*.
- **Payment:** Attributes: paymentID, customer, amount, date, mode, chequeNo. Methods: *record()*, *printReceipt()*.
- **CommissionReport:** Attributes: deliveryPerson, month, deliveredValue, commission. Methods: *calculate()*, *print()*.

3.8 Modes

The NAAS operates in the following modes:

1. **Normal Operation Mode:** Day-to-day data entry, subscription management, payment recording, and delivery list generation.
2. **End-of-Month Mode:** Triggered on the first day of each month. The system runs automated bill generation, outstanding due checks, reminder notice printing, and commission calculations.
3. **Report Mode:** The manager can generate on-demand reports at any time.
4. **Maintenance Mode:** Used by system administrators for database backup, publication price updates, and user management. End users cannot log in during this mode.

3.9 User Classes

User Class	Permissions and Responsibilities
Manager	Full system access. Can manage customers, subscriptions, publications, payments, delivery persons, and all reports. Responsible for monthly billing and commission processing.
Delivery Person	Restricted access. Can only view and print their own daily delivery list. Cannot access financial or subscription data.

User Class	Permissions and Responsibilities
Administrator	System-level access for configuration, user management, backup, and maintenance. Does not participate in day-to-day operations.

3.10 Concepts (Object/Class)

The core domain concepts of NAAS and their relationships are:

- A **Customer** is the central entity. A customer subscribes to publications, receives deliveries, gets billed, and makes payments.
- A **Subscription** is the contract between a customer and a publication. It has a lifecycle: *pending* → *active* → *suspended* → *cancelled*.
- A **Publication** is the product (newspaper or magazine) that is delivered.
- A **DeliveryPerson** is the agent who physically delivers publications to customers within an assigned zone.
- A **Bill** is the monthly financial document computed from delivery records.
- A **Payment** settles a bill partially or in full and triggers a receipt.
- A **DeliveryHold** temporarily suspends deliveries without cancelling the subscription.

3.11 Features

Feature ID	Feature Name	Description
F-01	Customer Registration	Register new customers with full contact and address details
F-02	Subscription Management	Subscribe, modify, and cancel publication subscriptions
F-03	Delivery Hold	Temporarily stop and resume deliveries
F-04	Delivery List Generation	Auto-generate sorted daily delivery lists per person
F-05	Monthly Billing	Auto-compute and print itemised monthly bills
F-06	Payment Recording	Record cash/cheque payments and print receipts

Feature ID	Feature Name	Description
F-07	Reminder Notices	Print polite reminders for overdue accounts
F-08	Auto-Suspension	Suspend subscriptions after 2 months of outstanding dues
F-09	Manager Reports	Monthly summary and detailed delivery reports
F-10	Commission Reports	Monthly commission computation and printing for delivery persons
F-11	User Access Control	Role-based login for manager and delivery persons
F-12	Data Backup	Daily automated database backup

3.12 Stimuli

The following table maps external stimuli (events) to system responses:

Stimulus ID	Stimulus / Event	System Response
S-01	Manager adds new customer	System creates customer record and assigns unique ID
S-02	Customer requests subscription	System validates publication and creates subscription entry
S-03	Customer requests modification	System checks advance notice; applies if valid, rejects with message if not
S-04	Customer requests delivery hold	System records hold dates; excludes customer from delivery lists during period
S-05	First day of month (scheduled)	System auto-generates bills for all active customers
S-06	Manager records payment	System updates balance, marks dues settled, prints receipt
S-07	Outstanding due > 1 month	System prints reminder notice for affected customer
S-08	Outstanding due > 2 months	System suspends subscription and notifies manager

Stimulus ID	Stimulus / Event	System Response
S-09	Manager requests delivery list	System generates sorted list for selected delivery person and date
S-10	Manager requests commission report	System calculates 2.5% commission and prints statement
S-11	Manager requests summary report	System generates current-month delivery and publication summary
S-12	Full payment received (suspended)	System reinstates suspended subscription

Chapter 4

Appendices

Appendix A: Sample Bill Format

NEWSPAPER AGENCY – MONTHLY BILL	
Customer Name:	Mr. Ramesh Kumar
Customer ID:	CUST-0042
Address:	12, Park Street, Kolkata – 700016
Billing Month:	March 2026
Publication Details:	

Publication	**Copies**	**Rate (INR)**	**Amount (INR)**
The Hindu (Daily)	31	5.00	155.00
India Today (Weekly)	4	35.00	140.00
		Sub-Total	295.00
		Outstanding (Feb 2026)	150.00
		Total Due	**445.00**

Appendix B: Sample Receipt Format

PAYMENT RECEIPT	
Receipt No.:	RCP-2026-0381
Date:	05/03/2026
Customer Name:	Mr. Ramesh Kumar
Customer ID:	CUST-0042
Amount Received:	INR 445.00
Mode of Payment:	Cheque
Cheque No.:	045621
Balance Due:	INR 0.00

Appendix C: Commission Calculation Example

Delivery Person: Rajiv Das **Month:** March 2026

Publication	Copies	Rate (INR)	Value (INR)
The Hindu	310	5.00	1,550.00
Times of India	248	5.50	1,364.00
India Today	40	35.00	1,400.00
	Total Delivered Value		4,314.00
	Commission @ 2.5%		**107.85**

Chapter 5

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